Andre Arcaina

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Education

Toronto Metropolitan University (Formerly Ryerson University) Sep 2022 – Apr 2027
Bachelor of Science (Honours), Computer Science (Co-op)
Coursework: Data Structures, Algorithms, Computer Networks, Databases, Operating Systems, Big Data, Distributed Systems

Experience

Linux Foundation (CNCF, LF AI & Data) Aug 2026 – Present
Open Source Contributor *Markham, ON (Remote)*

- Contributing to the development of cloud-native projects such as Flux CD, OpenTelemetry, gRPC-Go, Flyte, Kepler, and Trickster.
- Merged 7+ PRs across Go, Rust, and Python codebases, resolving bugs, enhancing Helm charts, streamlining configuration, improving test coverage, strengthening overall system reliability and maintainability.
- Expanded Flyte SDK's CI coverage to Windows by introducing an OS matrix, triaging 124 unit test failures/errors, fixing cross-platform compatibility issues, and explicitly skipping POSIX-dependent tests.

Environment and Climate Change Canada (ECCC) Jun 2025 – Aug 2026
Software Engineer Intern *North York, ON*

- Modernized 10+ mission-critical Java applications to Java 17 and Jakarta EE 10, reducing security risk and technical debt for national weather, maritime, and Coast Guard operations.
- Resolved 5+ production issues by investigating C++ decoder and Java service errors in Graylog, reducing incident triage time by 38% through Grafana and Prometheus monitoring.
- Built the first end-to-end test suite for a satellite decoding system in Go, integrating 25 automated scenarios into Jenkins CI.
- Refactored three C++ satellite data services, making failures easier to troubleshoot and improving overall system reliability.

Undergraduate Science Society of TMU (USSTM) Jan 2025 – Present
Software Engineer *Toronto, ON*

- Co-owned a contract-first Go REST API with 60+ routes for a campus item-lending service with 1850+ items, deployed and maintained on Amazon EC2 with Docker Compose and Caddy.
- Reduced regression risk by adding 40+ unit and integration tests, reaching roughly 70% coverage of core application packages.
- As one of two original engineers, reviewed 24+ PRs and onboarded 3+ frontend developers to the Go backend, Docker, Make, and OpenAPI workflow during a restructuring into a full-stack team.

DataKinetics May 2024 – Aug 2024
Software Engineer Intern *Ottawa, ON*

- Led development of a Spring Boot REST API and dashboard that converted COBOL Copybooks to JSON, letting engineers edit Db2 records and decode mainframe data for troubleshooting.
- Built 75 Postman requests and 97 assertions validating API responses, pagination, and error handling across three Db2 tables.
- Documented the project using Confluence for 12 R&D stakeholders, creating a shared reference for testing and onboarding.

Projects

Homelab Kubernetes Cluster | Linux, Virtualization, Networking, Observability, GitOps, CI/CD Blog Post

- Deployed a private cloud on Proxmox VE with a three-VM K3s cluster hosting personal media, public projects, and private services.
- Built a CI/CD GitOps pipeline using GitHub Actions and Flux CD to automate tests, container builds, and Kubernetes deployments.
- Implemented observability using Prometheus, Grafana, Loki, and Grafana Alloy on 11 services, with dashboards and alert rules for diagnosing infrastructure and application failures.

Distributed Key-Value (KV) Database | Go, Raft, BadgerDB, Docker, Kubernetes (kind) GitHub

- Designed and built a five-node distributed key-value database, focusing on replication, consensus, and node coordination.
- Containerized and deployed the cluster, then chaos-tested network partitions, leader failures, abrupt node crashes, and concurrent writes to verify fault tolerance and data consistency.

Skills

Languages: Go, Python, Java, C++, Rust, TypeScript, JavaScript, SQL, Shell (Bash)
Infrastructure: Kubernetes, Docker, Helm, Terraform, Linux, Cloudflare (Tunnel), AWS (EC2, S3, SES), GCP (Cloud Run, GCS)
DevOps/Tools: GitHub Actions, Jenkins CI, GitOps (Flux CD), Prometheus, Grafana, Loki, Graylog
Backend/Web: Spring Boot, FastAPI, React.js/Next.js, REST APIs, gRPC, WebSockets, PostgreSQL, Redis, NATS, Kafka